

The Noisy Work of Uncertainty Visualisation

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 The concept of signal suppression is referred to throughout, but it is really a combination of signal enhancement and signal suppression, and it would be helpful if this could be more clearly described, especially in the context of the examples.

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Abstract

Better representation of the uncertainty in a data visualisation is a focus of recent research activity. A problem with the current literature is that there is a lack of clarity about the definition of uncertainty and what it means to represent it in a plot. This confusion results in a significant amount of conflicting results in the literature, especially in experiments that assess the effectiveness of different uncertainty representations. In this review, we summarise the current literature, provide workable definitions, and illustrate these definitions with examples. In doing so, we ask what it really takes to achieve transparency in statistical graphics. It is hoped that it will be useful for guiding new graphics methodology and experimental research.



1. Introduction

What do we mean when we talk about “uncertainty visualisation”? The phrase can feel contradictory to anyone familiar with the term. Among statisticians, “uncertainty” is often discussed as an omnipresent spectre touching every stage of our analysis without ever being fully seen. Authors will often mention that the phrase is vague (Spiegelhalter 2017, Griethe & Schumann 2006), or avoid defining it by describing a list of things uncertainty *could* be (Kinkeldey et al. 2014, Hullman 2016), but rarely do authors attempt to discuss what uncertainty actually *is*. By contrast, visual statistics (information visualisations, data plots) are one of the most powerful tools in the statistician’s toolbox, allowing for quick and memorable communication that identifies quirks in our data that we didn’t even know to look for. We see this in datasets such as Anscombe’s quartet (Anscombe 1973) or the Datasaurus Dozen (Matejka & Fitzmaurice 2017, Locke & D’Agostino McGowan 2018), where visual statistics are able to highlight elements of the data that are invisible to the typical summary statistics. We also see this in recall experiments, where simply sketching a distribution before recalling statistics or making predictions can greatly increase the accuracy of those measures (Hullman et al. 2018, Goldstein & Rothschild 2014). ~~Taken together,~~ uncertainty visualisation implies a need to pull back the curtain and explore the unknowns of our analysis.



As nice as this sentiment is, it turns out to be easier said than done. Reviews on uncertainty visualisation rarely offer tried and tested rules ~~for effective uncertainty visualisation~~, instead commenting on the difficulties faced when trying to summarise the field. Kinkeldey et al. (2014) found most experimental methods to be ad hoc, with no commonly agreed upon methodology, formalisations, or a greater goal of describing general principles. Hullman (2016) noticed there is a serious noise issue in the field, with noise from participants misunderstanding visualisations, misinterpreting questions, and incorrectly applying heuristics, overwhelming any information we can glean from studies. MacEachren et al. (2005) identified so much contradicting evidence that they spent an entire page discussing the conflicting evidence for the question “Should I map uncertainty to colour hue?” Spiegelhalter (2017) concluded that different plots are good for different things, arguing against a universal best plot for all people and circumstances. Padilla et al. (2022b) summarised several cognitive effects that repeatedly arise in uncertainty visualisation experiments; these effects were each discussed in isolation as a list of considerations rather than an overarching theory for effective uncertainty visualisation.



“Science is built up of facts, as a house is built of stones; but an accumulation of facts is no more a science than a heap of stones is a house.” - Henri Poincaré (1905)

While these reviews are thorough in scope, none discuss how the existing literature contributes to the broader goal of uncertainty visualisation – that is, despite the wealth of reviews, the field of uncertainty visualisation remains a heap of stones. There is a mountain of work that identifies common heuristics found in uncertainty visualisations, evaluates competing plot designs, or starts a theoretical discussion on a niche aspect of the field. While important, each of these papers offers up its own bespoke motivation and methodology, with little reference to the uncertainty visualisation papers outside their fiefdom. The field is in desperate need of a unifying theory that can tie the conflicting and siloed research together. This review attempts to address this issue by offering a novel perspective on the uncertainty visualisation

problem. That is, we will use the wealth of established stone to construct a foundation to build a house.

2. The purpose of uncertainty visualisation

Mentions of “uncertainty visualisation” start springing up around 1990, across several different fields (Ibrekk & Morgan 1987, MacEachren 1992), each with its own motivation for the work. In computer science, the area appears to be motivated by issues in the public’s perception of random variables, with the hope that visualisations would give laypeople the ability to extract important information from graphical representations (Ibrekk & Morgan 1987). With similar concerns about the public’s understanding of randomness, the fields of psychology, statistics, and economics used “uncertainty visualisations” as a communication tool to mitigate the psychological bias associated with the communication of risk, a topic of concern since the early 1980s (Spiegelhalter 2017). In cartography, it was motivated by the inherent uncertainty of geoscience data, the practical use of visualisation as an exploratory tool, and the constrained visual channels from map representations (MacEachren 1992). These disparate motivations have blended together, and today, uncertainty visualisation is usually motivated by the vague goal of “decision-making”. This term has been used to mean the mitigation of psychological bias to ensure economically rational decisions (Padilla et al. 2022a, 2021, Kale et al. 2021), the facilitation of trust or confidence (Zhao et al. 2023, Yang et al. 2023), the ability to extract values related to a distribution (Sarma et al. 2023), the prevention of false discovery in plots (Sarma et al. 2024, Koonchanok et al. 2023), or the extraction of some other metric that is vaguely related to “uncertainty” (Chakraborty et al. 2024, Ndlovu et al. 2023). This gradual scope creep of the field, motivated by the hazy definition of “decision-making”, is the most likely culprit for the jumbled literature that makes up the field today.



Given that there is so much ~~subliminal~~ disagreement in uncertainty visualisation, how did these disparate motivations come to be seen as interchangeable? All discussions on uncertainty visualisation seem to have a common thread that connects them: the belief that the ultimate goal of uncertainty visualisation is not trust, rationality, or value extraction, but *transparency*. We see it said directly in reviews of the field (Padilla et al. 2022b), or when authors claim that failing to include uncertainty is akin to fraud or lying (Hullman 2020, Manski 2020). We see it when authors assert that uncertainty communicates the legitimacy (or illegitimacy) of the conclusion drawn from visual inference (Kale et al. 2018, Griethe & Schumann 2006). We see it when authors say uncertainty visualisations should communicate a degree of confidence (Correll et al. 2018, Boukhelifa et al. 2012) or validity (Hullman 2020, Griethe & Schumann 2006) in our conclusions. We see it when authors suggest uncertainty visualisation should “guide, qualify, or soften our judgements of uncertain data” (e.g. Wilkinson (2005) in his seminal work on the grammar of graphics). These authors are not wrong about the need for transparency in science communication: a six-month survey of anti-mask groups on Facebook during the COVID-19 pandemic showed that anti-maskers made persuasive arguments by exploiting inherent uncertainty ignored by pro-maskers (Lee et al. 2021).

Uncertainty visualisation is motivated by the need for a sort of “visual hypothesis test”, a sentiment expressed by some authors directly (Correll & Gleicher 2014, MacEachren 1992). A successful uncertainty visualisation would act as a “statistical hedge” for any inference we



make using the graphic. Since the purpose of a visualisation is to give a **quick gist** of the information (Spiegelhalter 2017), this hedging should be communicated visually without the need for complicated mental calculations. Therefore, an effective uncertainty visualisation should not just “show” uncertainty; untrustworthy conclusions should *not be visible*. If we refer to the conclusion we draw from a graphic as its signal, where uncertainty should make this signal harder to read as the “noise” increases, we can summarise the above information into three key requirements. A good uncertainty visualisation should:

- 1) Reinforce **justified** signals to encourage confidence in results.
- 2) Hide spurious signals that are overwhelmed by noise.
- 3) Perform tasks 1) and 2) in a way that is proportional to the level of confidence in those conclusions.



Usually, visualisations that are unconcerned with uncertainty have no issue showing justified signals, but struggle with the display of unjustified signals. Therefore, we **suggest calling this approach to uncertainty visualisation “signal-suppression”** since it primarily differentiates itself from the normal “noiseless” visualisation approach through criterion (2). This is the main criterion we will use to assess the current literature on uncertainty visualisation.



3. Current Approaches

3.1. Ignoring uncertainty

The most common way to visualise uncertainty is to simply not. A study conducted by Hullman (2020) found that only a quarter of authors surveyed included uncertainty in 50% or more of their visualisations, in part because authors are not sure how to calculate uncertainty. This is not entirely unreasonable, given that even visualisation authors themselves seem to be in conflict about what exactly uncertainty is. We will start with visualisations that ignore uncertainty with the hope that by looking at where uncertainty isn’t, we can better understand where it is.

3.1.1. What is uncertainty? It is surprisingly hard to describe what uncertainty is. Most authors avoid the problem and describe the many characteristics of uncertainty. Often, uncertainty is split by factors such as whether it is due to true randomness or a lack of knowledge (Begg et al. 2014, Spiegelhalter 2017, Gustafson & Rice 2019, Padilla et al. 2022b, Hullman 2016, Walker et al. 2003); quantifiable or unquantifiable (Spiegelhalter 2017, Walker et al. 2003, Padilla et al. 2022b); scientific or human (Benjamin & Budescu 2018, Gustafson & Rice 2019); systematic or random (Sanyal et al. 2009); statistical or bounded (Gschwandtner et al. 2016, Olston & Mackinlay 2002); accuracy or precision (Griethe & Schumann 2006, Benjamin & Budescu 2018, Hullman 2016), etc. There are enough qualitative descriptors of uncertainty to fill a paper, but none of this is particularly helpful in understanding how to integrate it into a visualisation.



Rather than trying to define uncertainty by looking at the myriad ways in which it *does* appear in an analysis, we may find it easier to look at where it *does not*. Descriptive statistics describe our sample as it is and summarise large data into a usable format, but they are not seen as the primary goal of modern statistics. In 19th-century England, *positivism* was the popular philosophical approach to science (**positivists** included famous statisticians such as



Francis Galton and Karl Pearson). Practitioners of the approach believed statistics ended with descriptive statistics, as science must be based on actual experience and observations (Otsuka 2023). In order to make statements about population statistics, future values, or new observations, we need to perform inference, which requires the assumption of the “uniformity of nature”, that is, we need to assume that unobserved phenomena should be similar to observed phenomena (Otsuka 2023). **Positivists believed referencing the unobservable was bad science, embracing descriptive statistics due to the inherent certainty associated with them.** Since uncertainty is nonexistent in descriptive statistics, it is clear that uncertainty is a by-product of inference: uncertainty is the noise that is both inseparable from our inference and meaningless without it.



Rather than extracting just one element of the distribution, if you can retain the whole distribution, that not only allows the uncertainty calculation to be reproduced, but also makes it possible to derive other estimates as well. ~~I think~~ it's also important to acknowledge that if you use this approach, you need to know the “chicken” (i.e. distribution) that gave birth to the “egg” (uncertainty estimate). In practice, uncertainties are sometimes calculated by other people or organisations, and the process used to derive them may not be known.

If we consider uncertainty to be a by-product of statistical inference, then uncertainty visualisations are the plots that depict an estimate, and therefore have an associated uncertainty. The most complete description of these estimates is their distributions. Rather than extracting just one element of the distribution, if you can retain the whole distribution, that not only allows the uncertainty calculation to be reproduced, but also makes it possible to derive other estimates as well. Suggesting distributions as a representation of uncertainty is not new. Kay (2023) originally suggested thinking about uncertainty visualisations as visualisations with distribution inputs, to replace the commonly used mean and standard deviation, removing the assumption of a Gaussian distribution. In practice, uncertainties are sometimes calculated by other people or organisations, and the process used to derive them may not be known. While some researchers believe ~~these~~ abstract notions of uncertainty, such as credibility (Thomson et al. 2005), forecaster confidence (Padilla et al. 2021), or uncertainty about uncertainty (Hadjimichael et al. 2024), are too complex to be quantified, this is not necessarily true. Abstract concepts such as human belief or credibility are regularly quantified by Bayesians, and hierarchical approaches are often used to model uncertainty about uncertainty.



3.1.2. Example: ignoring uncertainty. If visualising uncertainty is fundamentally visualising a set of random variables, what does “ignoring” uncertainty look like? Figure 1 shows plots of data from three different scenarios, each with either high or low uncertainty. The plots show the expected value of the input distribution. In plots R1 and R2, the expected value is the line from a simple linear regression on a car’s miles per gallon (mpg) and weight (wt) using the `mtcars` data, or a subset of it (available in `ggplot2` (Wickham 2010), and originally from Henderson & Velleman (1981)), with differing sample sizes creating differing levels of uncertainty. Plots S1 and S2 show a choropleth map of Iowa, where counties are coloured according to a simulated temperature measurement, with measurement error being the uncertainty associated with the measuring instrument. Plots G1 and G2 show samples simulated from five populations (A-E). In G1, distributions have the same low variance, and in G2, they have the same high variance. In the linear regression, we see a downward trend for both levels of uncertainty. Here, we elected to plot the data under the fit, to provide

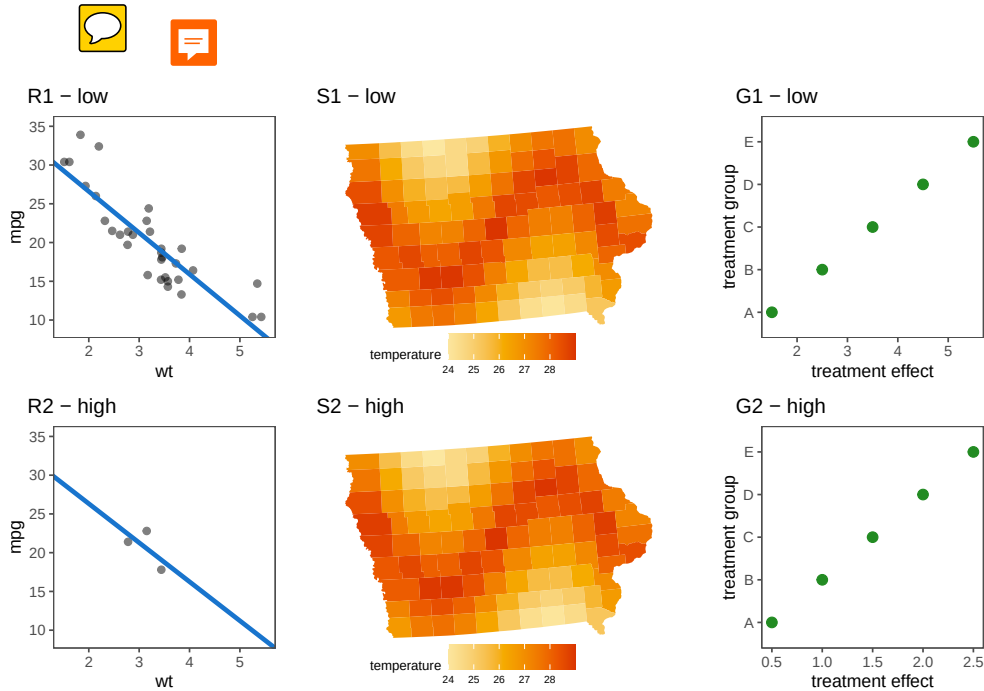


Figure 1: Three example types of data and associated plots that will be used to illustrate various choices of uncertainty representation throughout the paper: scatterplot and regression line (R1, R2), spatial choropleth map (S1, S2), grouped dotplot (G1, G2). Two levels of uncertainty (low, high) are used with each example. Ignoring the points in R1 and R2, there are no differences between the high and low uncertainty versions. Ignoring uncertainty can lead to misrepresentation of data.

context for this example, and help follow the thinking through to the later illustrations. Often, only the fitted line is shown. In the map, you should see a sine wave spatial trend, and in the univariate distributions, an incremental increase in treatment. If we were to ask a reader, “Can you see a difference between the plots in the top row versus the bottom row? Is the strength of the trend communicated through the visualisation?” The answer to both of these questions will be no for S1, S2, G1, and G2, as the high and low variance cases are identical. For R1 and R2, the answer would likely only relate to the points in the plots, not the regression line.

These examples will serve as touchstones for a discussion of uncertainty visualisation, focusing on the approaches suggested in the literature.

3.2. Uncertainty as a statistic

Uncertainty is also often treated as another statistic, for example, the exceedance probability map of spatial data (see Kuhnert et al. (2018) with software available in Lucchesi et al. (2021)), where the probability of exceeding a specified value is displayed on the map, placing the focus on extreme events. For risk communication, Spiegelhalter (2017) recommends showing probability with a set of coloured icons to communicate the relative number of affected individuals. The summary plot (Potter et al. 2010) was also developed to address concerns about uncertainty in data displays. This dense display combines the plot of a single

set of numeric values with a density estimate, boxplot, and statistical moments.

Some authors take this approach because they explicitly believe uncertainty is a variable of importance (Blenkinsop et al. 2000), while others straddle the line, asserting uncertainty is acting as signal and noise, and should fulfil both roles (Peña-Araya et al. 2025). Simply put, these approaches might be described as swapping out the statistic for an “uncertainty statistic” to get an “uncertainty visualisation”. Is it really that easy?

3.2.1. Example: visualising variance. Figure 2 depicts the six plots showing this approach for the data introduced in Figure 1. The original central value estimate has been replaced with an uncertainty statistic. Plots R1 and R2 show the residual plot of our linear regression instead of the scatterplot and the regression line. The visual patterns we are looking for in this plot are distinct from the linear regression, so it is hard to evaluate it relative to the trend. Sometimes the variance approach is still related to our original display, as we can see in the exceedance probability maps depicted in plots S1 and S2. These map $P(\text{temperature} > 27)$ to colour for each county. The sine wave trend is clearly visible when the error is low, but barely visible with high error. Our visualisation of the univariate groups does reveal something interesting: the variance is constant in G1 but different for each group in G2. It is a nonsensical display, though, because the trend has completely disappeared. In practice, two plots are typically presented: one showing the main estimate and the other showing the uncertainty.

3.2.2. What is an uncertainty visualisation, then? What an uncertainty visualisation is or is not is one of the most pervasive divides in the literature. For example, Wilkinson (2005) mentions that popular graphics, such as pie charts and bar charts, omit uncertainty. Wickham & Hofmann (2011) suggests their product plot framework, for area plots like bar charts and also histograms, needs to be extended to include uncertainty representation. However, pie charts, bar charts and histograms have all been used in a significant number of experiments as examples of an “uncertainty visualisation” (Ibrekk & Morgan 1987, Olston & Mackinlay 2002, Zhao et al. 2023, Hofmann et al. 2012). What is going on here?

This conflict stems from a subconscious disagreement about the purpose of uncertainty visualisation. If you believe uncertainty visualisation is about communicating risks or random variables, uncertainty visualisations are just visualisations of “uncertainty statistics” or distributions. On the other hand, if you believe uncertainty visualisation is about suppressing false signals visually, then you see an uncertainty visualisation as a transformation of an existing graphic that adds the uncertainty in. The former has no limitation on the visual appearance of an “uncertainty visualisation”, allowing pie charts, bar charts or histograms, so long as the graphic is visualising “uncertainty”, while the latter believes uncertainty visualisations only exist in relation to some “normal” visualisation. When we refer to the graphics depicted in Figure 2 as “uncertainty visualisations”, we are classifying visualisations by the data they display, not their visual features. This is not the standard approach in statistical graphics. A scatter plot that compares means and a scatter plot that compares variances are both scatter plots.

Unlike plots, which are not defined by a statistic, uncertainty is *only* defined in relation to our visual statistic, a topic that frequently appears in the literature to be dependent on the “goals” of our analysis. Meng (2014) commented that what is kept as data and what

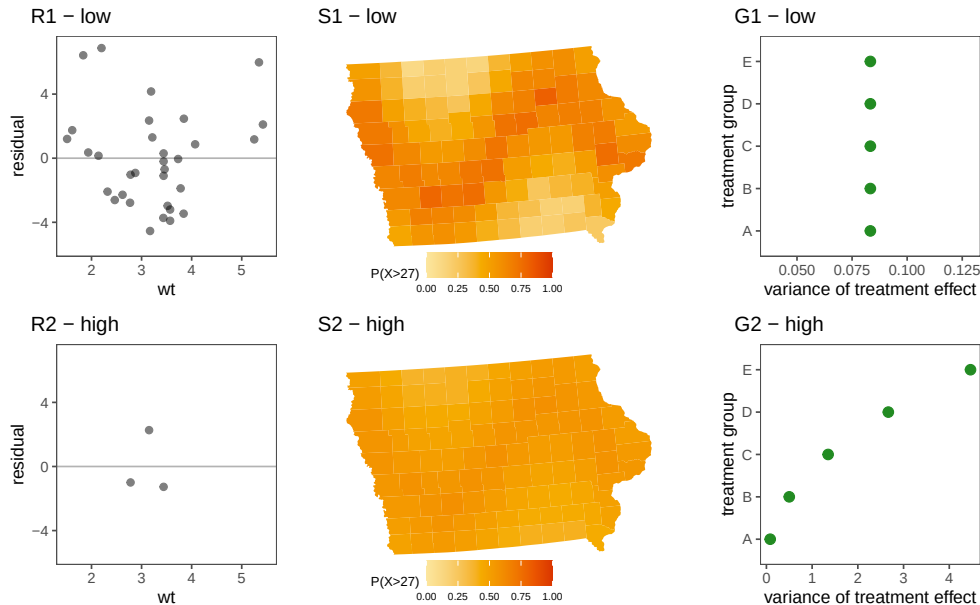


Figure 2: Treating the uncertainty as a statistic, with the same six examples. The regression (R1 and R2) is a scatterplot of residuals vs explanatory variable, separated from the regression making comparison of uncertainty related to the trend more difficult. For the choropleth (S1, S2): instead of temperature, the probability of exceeding 27°C is shown. This has the effect of highlighting (sine wave) trend in the low error data, and de-emphasising it in the high error data. The plots G1 and G2 have replaced the treatment effect with the variance on our treatment effect. It is a bit nonsensical, but we learn something interesting that was not seen earlier: the variance in G2 is not uniform like that in G1.

is tossed away is determined by the motivation of an analysis - what was previously noise can become signal depending on the question. Otsuka (2023) suggested that the process of observing data to calculate statistics is largely dependent on our goals, because the process of boiling real-world entities down into probabilities depends on the relationships we seek to identify within our data. Wallsten et al. (1997) argue that the best method for evaluating or combining subjective probabilities depends on the uncertainty the decision-maker wants to represent, and why it matters. Fischhoff & Davis (2014) suggested we should have methods for communicating uncertainty depending on what the user is supposed to do with it. Spiegelhalter (2017) says we “cannot assess the quality of risk communication unless the objectives are clear”. Peña-Araya et al. (2025) asserted that whether or not uncertainty is a source of doubt depends on the context. The sentiment behind this repeated point is clear: the role of uncertainty or signal is not dependent on the “type” of statistic, on the source of the information, or the methods we use; it is determined by the statistic we wish to draw inference on. Therefore, the fundamental problem with the “uncertainty statistic” approach is that the uncertainty in the plot isn’t acting as noise; it is acting as signal.

If the uncertainty in a graphic is acting as a signal, there isn’t an interesting perceptual challenge associated with the visualisation: the uncertainty can be displayed using standard

principles of graphic design. In changing the inferential statistic, we also haven't dealt with the original problem of integrating noise, as these "uncertainty statistics" *also have associated uncertainty in the estimates* (e.g. variance of standard deviation estimate) that is being ignored. There is nothing wrong with explicitly visualising variance, error, bias, or any other statistic. These metrics provide important and useful information for analysis and decisions. The problem with this approach is that it means everything is an uncertainty visualisation, and if everything is an uncertainty visualisation, nothing is.



3.3. Uncertainty as a variable

Another common characterisation of uncertainty is as just another variable to be integrated into the visualisation, which means uncertainty visualisation is, at its core, a high-dimensional visualisation problem (Griethe & Schumann 2006, Peña-Araya et al. 2025). This occurs within computer science (Kinkeldey et al. 2014), cartography (MacEachren et al. 2005), and statistical graphics (Wilkinson 2005).



Discussion of the visualisations focuses on how "integrated" the uncertainty is with the estimate. Kinkeldey et al. (2014) identified a split between intrinsic plots, where we map uncertainty to the colour or size of the geometric object of our estimate, and extrinsic plots, where uncertainty is mapped to a separate geometric object, such as glyphs or error bars. Similarly, Padilla et al. (2022b) classified uncertainty visualisations as graphical annotations (extrinsic), and probability mapped to a visual encoding channel (intrinsic), or a hybrid of the two. It is unclear if these levels of "integration" in a plot design affect its ability to suppress signals.



3.3.1. Example: mapping two independent variables. Figure 3 shows the six examples introduced in Figure 1. Here, uncertainty is mapped to a spare aesthetic in the plot. In the grammar of graphics, variables are mapped to aesthetics, like position, colour, and size, within a plot. In plots R1 and R2, the standard error of the slope is represented by the width of the line. This is common, but it fails to represent the standard error of the intercept alongside the slope. We might think we can examine the width of the line at `wt=0`, but this would be incorrect. Here, it has been included, less obviously, by mapping the standard error of the intercept to transparency. While these help to give a gestalt of the uncertainty, they are not exacting representations. One would expect that the width above and below the line is one standard error, but why not map the width to two standard deviations, or even three? Interpreting transparency into a numerical quantity is also virtually impossible, so using this aesthetic mapping is effectively useless.



A bivariate colour palette map is shown in plots S1 and S2. With two dimensions of the plot reserved for spatial position, it is difficult to incorporate the error. The bivariate colour palette maps the estimate to hue and the error to saturation, keeping the signal and noise contained to one visual aesthetic. This has the unfortunate effect of making the signal appear stronger when the error is higher (S2). Colour perception is a wild beast that is hard to tame.

The extrinsic approach, shown in plots G1 and G2, has the estimate represented by a point, and uncertainty computed as a 95% confidence interval mapped to line length. The trend is still visible in both displays. It could be argued that where the variance is high, the trend remains a main focus; that is, the display fails to sufficiently suppress the signal.



Because all of these graphics visualise the distribution's estimate and uncertainty as two separate pieces of information, the message of the plot is “here is the trend *and* here is the uncertainty”. It is worthwhile to examine why this occurs, to see if we can move towards a version of this plot where we are able to communicate signal and noise simultaneously.

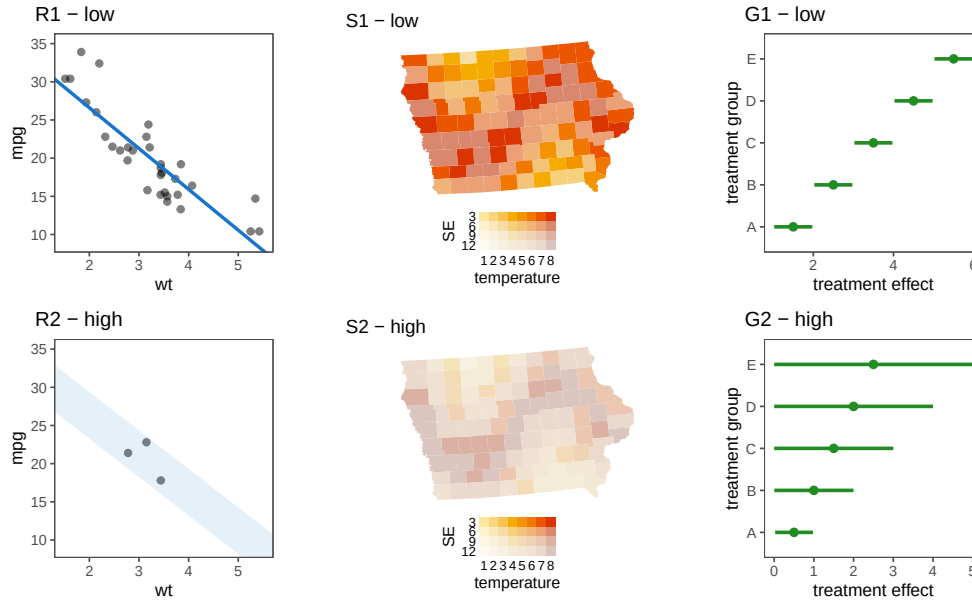
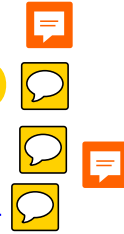


Figure 3: Treating the uncertainty as a variable, with the same six examples. In plots R1 and R2, the standard error of the slope is mapped to the line width. The standard error of the intercept is mapped to the transparency, which is less conventional. In plots S1 and S2, a bivariate colour palette is used with mean mapped to the hue, and error mapped to saturation. Plots G1 and G2 represent the mean of each group as a point, and the variance as an interval. The uncertainty is integrated well with the signal, but for the choropleth, the result is undesirable: the signal is easier to see when the error is higher.

3.3.2. Can we visualise a “single integrated uncertain value”? The reality is, based on our discussion on inferential statistics, uncertainty *isn't* a separate variable: it is a component of the random variable that is indistinguishable from the random variable itself. Similarities between the “as a variable” approach and the “as a statistic” approach are apparent when we read motivations for visualising an estimate (i.e. Figure 1) and variance (i.e. Figure 2) side-by-side using two separate graphics. This organisation is vulnerable to change blindness (Simons & Levin 1997) as one needs to switch focus between two displays.) The preference for the methods utilised in Figure 3 is usually motivated by the difficulties in combining information on two separate graphics (Moritz et al. 2017, Correll et al. 2018), rather than an understanding that the “uncertainty statistic” approach is not philosophically sound. To achieve signal suppression, we need to visualise noise and signal together as a “single integrated uncertain value” (Kinkeldey et al. 2014) rather than as two separate statistics.

Just because we can still see the signal in Figure 3, that does not mean the reading of the estimate is completely independent of the uncertainty. When making any visualisations, we

usually want the visual channels to be separable, that is, we don't want the data represented through one visual channel to interfere with the others (Smart & Szafir 2019). It is also interesting to know whether readers can see all structures when there are multiple structures present in a data plot, or whether they systematically fixate on one (VanderPlas & Hofmann 2017). Separability may be desirable in standard data visualisation, but in uncertainty visualisation, it allows the estimate and its variance to be read independently, potentially leading to the uncertainty being ignored (Padilla et al. 2022b). Therefore, rather than trying to maintain visual separability, the goals of uncertainty visualisation align far better with the pursuit of visual integration. In an ideal system, our estimate and uncertainty would be manipulated separately, but would be so *well-integrated that they are read as a single channel by the human brain*. The problem is that even if we can implement **the most extreme versions of integrable**, our methods fall short, as illustrated by the bivariate colour palette map in Figure 3. Colour hue and brightness are ~~one of the~~ classic examples of integrable variables (Vanderplas et al. 2020), and decreasing saturation should make the colours harder to distinguish, but the signal is still **clearly visible with the high variance**. ~~This is to say nothing of the fact that~~ multi-dimensional colour palettes can make the graphics harder to read and less accessible (VanderPlas & Hofmann 2015).



3.3.3. Another example: mapping combined variables. The Value Suppressing Uncertainty Palette (VSUP) (Correll et al. 2018) was designed with the intention of preventing high uncertainty values from being extracted from a map by blending colours together as they become less certain. Figure 4 shows the spatial example (S1, S2) using the VSUP approach. Since the palette was designed with the extraction of individual values in mind and it has only been tested on simple value extraction tasks (Correll et al. 2018) or search tasks (Ndlovu et al. 2023), we can see that, at least for our example, when the uncertainty is high, the spatial trend has functionally disappeared.

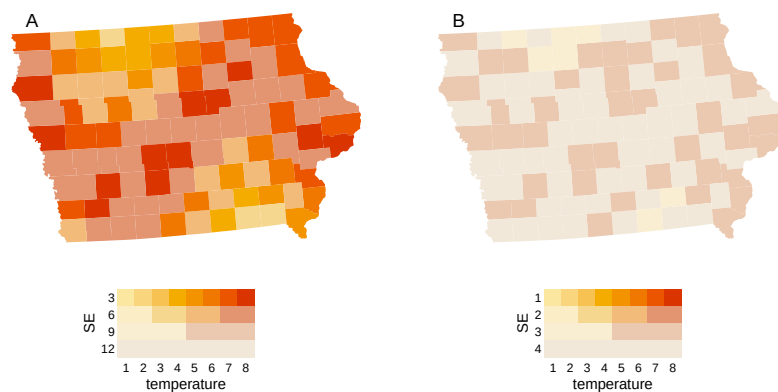


Figure 4: The spatial examples displayed with a choropleth map using a VSUP colour palette, where hue is blended when ~~increased uncertainty~~. Plot B has successfully produced signal suppression. Look closely at the scales, though: it may have another explanation.

Finally, we have signal suppression! Well, not really, sorry, we tricked you. The two plots depicted in Figure 4 actually show the exact same data; they are both the low variance case. If you look closely, you can see that the two plots have a different scale, where plot A has been scaled **according to our existing knowledge about this data**, while plot B



scaled using the range of the data passed to the plot. Since the variance and estimate are scaled independently, arbitrary differences in **the range of our variance**, unrelated to the estimate itself, will have significant impacts on the   appearance of our plot. The scale issue in VSUP maps was also recognised by [Kay \(2019\)](#), who noted that the suppression of any one hypothesis largely depends on the methods we use to combine the palette, and the variance levels at which the blending occurs. This means that, for us to know that our plot will successfully perform signal suppression, we need to already know what signal we are trying to suppress and set up the VSUP palette accordingly. This means that VSUP maps are not suitable for exploratory data analysis.

3.3.4. Uncertainty and exploratory data analysis. The lack of uncertainty in descriptive statistics is due to the lack of inference. Descriptive statistics are actually a small piece of a much larger field, exploratory data analysis (EDA), that tends ~~not to perform~~ statistical inference. [Tukey et al. \(1977\)](#) described EDA as the process of searching for interesting hypotheses (“the greatest value of a picture is when it forces us to notice what we never expected to see”), and defined it in relation to confirmatory data analysis (CDA), the process of verifying a hypothesis. ~~There are more~~ subfields of EDA: initial data analysis ([Huebner et al. 2016](#), [Chatfield 1985](#)), which involves checking assumptions and data quality prior to CDA, and model diagnostics (e.g. [Belsley et al. \(1980\)](#)), including posterior checks of model fit. What binds these pursuits together is their reliance on visual summaries for making assessments and an absence of formal inference.

[Hullman & Gelman \(2021\)](#) argued that the EDA and CDA are not entirely distinct, as it is often difficult to draw a hard line. Our belief, ~~as with many concepts,~~ is that these ~~approaches~~ exist on a continuum, where we have an inherent trade-off between the number of hypotheses we can look for and the certainty of any conclusions reached. It can help to think of the knowledge-generating process of EDA and CDA as the nozzle on a hose with multiple spray options, where EDA is a fine misting spray that touches everything in the room, and CDA is a high-pressure jet capable of obliterating any and all debris from any single spot.

Viewing EDA and CDA as a dichotomy can create some confusion when it comes to understanding the source of uncertainty in our analysis. This is why we have avoided the topic until now, despite the fact that an uncertainty visualisation system for EDA is one of the most discussed topics in the field ([Hadjimichael et al. 2024](#), [Peña-Araya et al. 2025](#), [MacEachren et al. 2005](#), [Sarma et al. 2024](#), [Griethe & Schumann 2006](#)). The EDA versus CDA dichotomy can be compared to the dichotomy between induction, for building theories, and deduction, for testing theories, from formal logic. One of the trade-offs in the two methods is that deductive conclusions provide certainty, while conclusions from induction are inherently uncertain. This means that EDA, the inductive counterpart, has uncertainty in any conclusions reached, with the requirement to follow up our newfound hypothesis with CDA if we want true certainty. This adds another layer of confusion to the study of uncertainty visualisation: conflating the uncertainty in our data with the uncertainty that is inherent to an exploratory process (EDA).

Authors often over-compensate for the inherent EDA uncertainty, pre-emptively hedging against every false inference that could possibly be drawn from a graphic. [Hullman & Gelman \(2021\)](#) argues there is no such thing as a “model-free” visualisation. [Guo et al.](#)

(2025) provides a grammar for visualising statistical model checks. The lineup protocol (Buja et al. 2009) provides the viewer with plots of the data in a field of plots of null data where any patterns seen are due to sampling variability. The Rorschach protocol, from the same paper, shows only null plots to give the reader some intuition for what spurious sampling patterns exist. Savvides et al. (2019) provides a statistical super-test against multiple comparisons driven by probabilistic arguments. There is a CDA quality to these approaches. The sum total of a lot of CDA is not EDA, just as swinging a high-pressure jet around a room is not equivalent to using a misting spray. While EDA and CDA may be along a continuum, we cannot simultaneously perform EDA and CDA as the approaches are, philosophically speaking, perpendicular to one another.  

To truly create an uncertainty visualisation approach that is capable of EDA, we need to accept that uncertainty is inherent to the method, and it cannot be pre-emptively removed from the visualisations. If we accept this fact, then the only possible source of uncertainty in an uncertainty visualisation system that performs EDA is from the data itself. That is to say, the only possible way for there to be uncertainty in a visualisation designed for EDA is that the data itself represents inference that has been done earlier in our analysis. We can see this in the original description of Figure 1, where the uncertainty in all cases, even measurement error, represents inference that was performed earlier in our analysis.   

In the VSUP approach, our ability to arbitrarily decide which values to blend at, or which suppression approach to use, means that the “uncertainty” we are visualising will be informed by the conclusions we are drawing, and not a product of the data itself – an antithetical approach to EDA. For a visualisation to be suitable for EDA, it should always look the same regardless of what hypothesis we plan to draw from it. As much of the transparency in data visualisation comes from this feature in EDA, it is reasonable to set it as a requirement of our visualisations. Ensuring this property means we cannot treat signal and noise as separate variables, but rather as a single integrated unit. 

3.4. Uncertainty as a distribution

Rather than trying to reduce random variables to a single value or pair of values, why not visualise the whole distribution? This approach is found in computer science’s hypothetical outcome plots (HOPs), which animate a sequence of potential outcomes of a distribution (Hullman et al. 2015), in geoscience’s pixel-maps (Lucchesi et al. 2021, Blenkinsop et al. 2000), and in statistics multiple forecasts (Hyndman & Athanasopoulos 2021).  

3.4.1. Example: visualising samples. Figure 3 shows the six examples introduced in Figure 1, shown as distributions. The distribution is represented by either a sample of outcomes or a quantile dot plot. Plots R1 and R2 show a linear regression as a sample of possible outcomes from the distribution. The distribution used for both the slope and intercept is normal with the conventional mean and standard error. Plots S1 and S2 show a pixel map, which is a choropleth map where each county is coloured by a sample of outcomes from the distribution. Plots G1 and G2 show each univariate distribution as a quantile dot plot. We can see that the strong downward trend in the linear regression, the sine wave in the choropleth map, and the incremental increase in the univariate distributions are all clearly visible in the low variance case, but disappear in the high variance cases. The graphics have achieved signal suppression. Visualising the random variable as a distribution gives additional information,     

such as the previously **hidden bimodality** of the univariate distributions in G1 and G2.

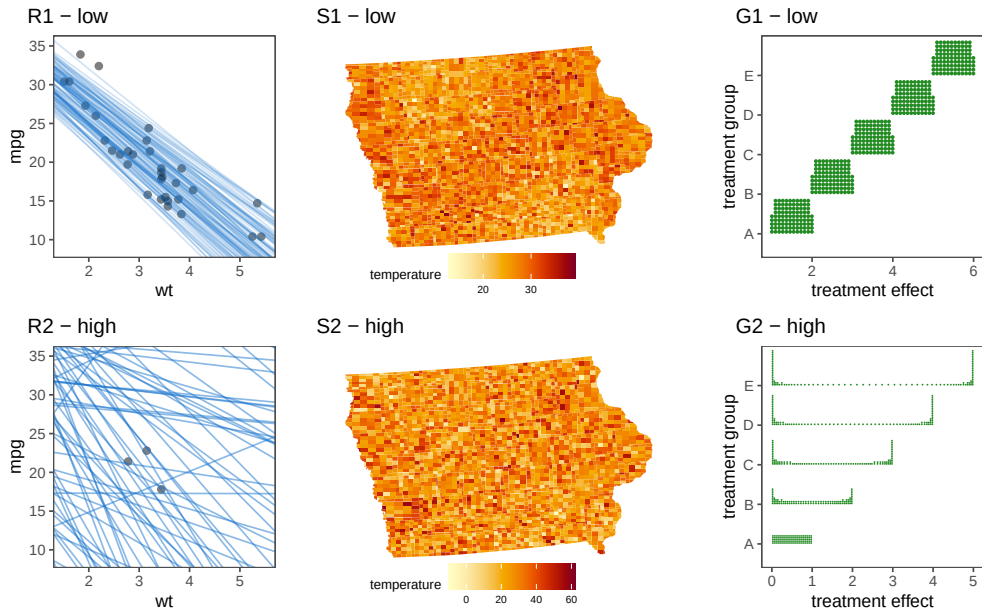


Figure 5: Treating **the uncertainty as a distribution**, with the same six examples. Plots R1 and R2 show a linear regression as a sample of possible outcomes from the distribution. Plots S1 and S2 show a pixel map. Plots G1 and G2 show the groups as quantile dot plots. In each case, the signal (regression line, sine wave, increasing trend) has disappeared with high uncertainty.

3.4.2. Quantified versus unquantified uncertainty. By showing our distribution as “data”, we are able to read the “uncertainty” plots using the same perceptual mechanisms we use to read the “non-uncertainty” plot. This should lead to more effective communication, as people tend to read more complicated visualisations like the bivariate and VSUP plots the same way they read the simple choropleth counterpart (Ndlovu et al. 2023). This approach also does not significantly hinder our ability to extract the individual values mapped by the previous plots, as extracting global statistics from a sample can be done with relative ease (Franconeri 2021). We are able to include more information by offloading more computation to visual processing, but what is the limitation on this approach? Which aspects of uncertainty should be processed by our visual system, and which should be processed by our statistical computation? ~~It is not obvious from the question, but~~ this is actually a question about how much of our uncertainty should be quantified.

Quantified uncertainty usually focuses narrowly on concepts such as probability, confidence intervals, variance, error, or precision (Hullman et al. 2018, Maceachren et al. 2012, Thomson et al. 2005), while unquantified uncertainty often includes a broader range of concepts like missing values, reliability, model validity, or source integrity (Griethe & Schumann 2006, Wilkinson 2005, Pang et al. 1997, Pham et al. 2009, Boukhelifa et al. 2017). When discussing uncertainty, we typically ~~include~~ these unquantified uncertainties, not because these things *are* uncertainty, but because they can *create* uncertainty when we perform inference. This is

often because these unquantified uncertainties violate our assumptions of the uniformity of nature (Otsuka 2023).



Sometimes we are able to visualise these assumption violations directly. For example, we can check for structure in our missing data using the `naniar` package Tierney & Cook (2023) that allows us to include missing values as a “shadow” alongside our usual visualisations. This approach amounts to just “showing the data”, which is a simple but effective option for uncertainty visualisation that is largely overlooked. While this approach is useful for better understanding data, it will not eliminate trends that have become invalid due to structure in missing data or an invalid model. We can only integrate uncertainty as noise when that uncertainty has been *quantified* as an effect on the estimates we are visualising. This is not to say one method is preferable; visualising both quantified and unquantified uncertainty is necessary for a healthy analysis. Data analysis often works in cycles, where we find assumption violations using EDA, quantify the effect of these violations on inference, and then visualise the output of that inference using uncertainty visualisation.



4. Evaluating uncertainty visualisations

Unfortunately, the conflicting results in the field are not limited to plot design and extend to the experimental findings as well (MacEachren et al. 2005, Kinkeldey et al. 2014, Hullman 2016). There are as many explanations for the noisy evaluation studies as there are contradictions in the research itself. Kim et al. (2019) believes there is some interference in results from participants' prior beliefs; Hullman (2016) believes the noise in the literature could come from visual heuristics, subjective probabilities, unknown participant utility functions, or a misunderstanding of statistical concepts (such as confidence intervals). Kinkeldey et al. (2014) suggest the perception of visualisation changes by audience, so we cannot expect the same results between different subpopulations. Brennen & Tuerk (2018) attributes evaluation difficulties to cognitive load from complicated uncertainty visualisations, as well as the participants' prior experience in the topic. While these issues will certainly have some impact on our ability to synthesise, none of them is unique to uncertainty visualisation. Rather, the issue is likely due to a disconnect between the evaluation methods used and the stated goals of each experiment, a common issue in visualisation evaluation experiments (Vanderplas et al. 2020).



4.1. Current evaluation methods

4.1.1. Value extraction. Uncertainty visualisations are most commonly evaluated based on how accurately viewers can extract an estimate and its variance (Kinkeldey et al. 2014, Hullman et al. 2019). This is not unusual, as direct observation is the simplest way to verify that information can be accurately read from a graph (Vanderplas et al. 2020). Unfortunately, this approach doesn't work for uncertainty visualisation. The second we ask a specific question about a statistic, that statistic becomes inferential, even if the plot was not the intent behind the question. By shifting the focus from $\hat{X}$ to $Var(\hat{X})$ or $P(\hat{X} < x)$, we end up evaluating visualisations on their ability to convey uncertainty statistics, rather than their ability to perform signal suppression. Even if the authors do not realise it themselves, there is nothing unique to uncertainty in these studies, so when we boil the findings down to generalised results, they simply restate existing principles within information visualisation. Some of the findings are obvious: participants were more accurate when reading a probability





expressed as text than when they had to extract it from a graphic (Cheong et al. 2016, Savelli & Joslyn 2013). Other studies replicate existing research, such as the finding that a probability mapped to a position is more accurate than one mapped to an area (Ibrekk & Morgan 1987, Gschwandtner et al. 2016), established as part of the hierarchy of perceptual tasks more than 40 years ago (Cleveland & McGill 1984; replicated by Heer & Bostock 2010). This extends beyond simple accuracy evaluations: Sanyal et al. (2009) found that colour was more effective than size when searching for extrema in variances; we have known that pre-attentive aesthetics, such as colour, are more efficient for search tasks since the 1980s (Vanderplas et al. 2020). By classifying these studies as evaluations of “uncertainty visualisation” while evaluating uncertainty as a signal, we are encouraged to see successful examples of signal suppression as failure. This approach leads authors to advise against particular aesthetic mappings for uncertainty, because they cause participants to have more difficulty extracting values (Blenkinsop et al. 2000). This conclusion is antithetical to the goals of signal suppression and occurs because these methods evaluate uncertainty as a signal, not as noise.

4.1.2. Trust, confidence, and risk aversion. If we cannot directly measure uncertainty for fear that it turns into a signal, we might then assume we can measure the secondary benefits of increased transparency. This seems to be the approach of many visualisation authors, as secondary benefits such as trust, confidence, and risk aversion are all frequently used in uncertainty evaluation studies (Hullman et al. 2019). Unfortunately, measuring these secondary effects often leads to confusing conclusions that simultaneously argue for and against the inclusion of uncertainty.

This is most commonly noticed in the use of trust as a measure, as several authors have commented that measuring trust, and not transparency, can lead to a questionable subtext that argues against transparency (Spiegelhalter 2017, O’Neill 2018). We see this directly play out in the visualisation literature, where surveyed visualisation authors explicitly said they didn’t include uncertainty due to the fact that they might decrease trust in their conclusions (Hullman 2020). This sentiment is also true for confidence, as Blenkinsop et al. (2000) commented that visually integrable depictions of uncertainty should be avoided, as they decrease the viewer’s confidence in their extracted data values.

Another secondary effect that is similar to trust and confidence is risk aversion. Risk aversion is an economic term used to describe an agent who would choose a random variable with a lower expected payout because it also has a lower variance. Risk aversion’s role in the uncertainty visualisation is unclear, as authors will argue uncertainty should elicit more risk aversion in one paper (Hullman et al. 2019), and argue for less risk aversion (by proxy of suggesting rational agents as a benchmark) in the next (Wu et al. 2023). Ultimately, these approaches have similar issues to value extraction studies, except they are slightly more confusing in their goals, leading them to simultaneously argue for and against the inclusion of uncertainty in a visualisation.

4.1.3. Alternative approaches. Often, authors understand that the effects of uncertainty are more complicated than simple value extraction. These studies indicate that accurately capturing uncertainty will be more complicated than simply avoiding value extraction or trust as a measure.

One approach is to ask indeterminate questions, such as asking participants for the “best estimate” (Ibrekk & Morgan 1987), or to select which distribution is the “furthest to the right” in a lineup (Hofmann et al. 2012). In both cases, the ground truth is based on the mean of the distribution, which is not as indeterminate as the question. This approach can lead to inconclusive results, as we are left unclear whether it was the phrasing of the question or the plot design that caused the participants to answer incorrectly.

On the other hand, questions that are incredibly specific about the distribution information can confuse the participants and induce noisy results. For example, Hullman et al. (2015) asked participants to compare two normally distributed groups, A and B, and had many participants say that group A was more likely to be bigger, despite group B having a higher mean. Gschwandtner et al. (2016) asked participants the “probability that the interval has already ended at the marked point in time?” and participants replied with the probability that the interval had already started.

The confusion around trying to capture the effects of uncertainty can also (understandably) extend to the authors of the study itself. We can see an example of this in Padilla et al. (2017). In order to answer the question correctly, the first experiment required participants to assume that an oil rig being “more likely to be hit” by a hurricane would *not* translate to the rig sustaining “more damage”. The second experiment required participants to assume the opposite.



4.1.4. Effective methods. This is not to say all evaluation studies fail to properly evaluate uncertainty as noise. There are several studies that ask participants to identify a particular signal that the noise is trying to obfuscate (Kale et al. 2018, Correll & Gleicher 2014), which seems to be an effective method. The only problem with these studies is that most uncertainty visualisation methods are not integrated into *the grammar of graphics* (Wilkinson 2005, Wickham 2010), so we regularly see comparisons between disparate plots that would never be considered substitutes for one another outside the artificial “uncertainty visualisation” framework they are placed within. For example, Kale et al. (2018) compared static bar charts with error bars to a bar chart with animated samples, meaning that any difference in participants’ ability to read the plot could be due to the statistic (confidence interval versus sample), the geometry (bars versus intervals), or the use of animation (static versus animated plots). This issue was rectified in their second experiment, where they compared overlaid and animated samples, giving us an insight into the types of visualisations that are appropriate to compare in uncertainty visualisation experiments. This means that even when evaluations *are* done correctly, there is no generalisable theory we can take from the results.

The point here is not to accuse the authors of poor academic rigour. The papers are (usually) logically consistent and well-formulated pieces of work. Rather, the point is to illustrate that evaluating uncertainty as noise is surprisingly difficult. Designing tests for signal suppression will require a formalisation of uncertainty within the grammar of graphics, as well as improved evaluation methods.

4.2. Implicit Hypothesis Testing

The main problem with current uncertainty visualisation evaluations is that they often require explicit (or convoluted) questions about the variance. Asking direct questions about

 The concept of signal suppression is referred to throughout, but it is really a combination of signal enhancement and signal suppression, and it would be helpful if this could be more clearly described, especially in the context of the examples.

the statistics or outcomes is not an explicit requirement of visualisation evaluations. In their review of testing statistical graphics, Vanderplas et al. (2020) drew a distinction between explicit tests, where participants are asked direct questions about specific features of a plot, and implicit testing, where users identify both the purpose and function of the plot. The lineup protocol is the most salient example of the implicit approach. Lineups are a confirmatory visualisation tool where participants are shown a set of M plots, and asked to identify the plot that is the “most different”, leaving participants to decide what “most different” means to them, even if it is not what the authors intended (VanderPlas & Hofmann 2017). The implicit test does not limit the versatility of the approach, with the lineup being used to evaluate the effectiveness of different types of plots (Hofmann et al. 2012), colour palettes (Reda & Szafr 2021), and design decisions (VanderPlas & Hofmann 2017).

Lineup protocols are not only useful for implicit testing: they also have parallels to hypothesis testing that can be leveraged in uncertainty visualisation. The concept of signal suppression is, at its core, an assertion of statistical validity: the visibility of signals should be directly proportional to p -values or some equivalent measure. This comparison is not new in uncertainty visualisation, where parallels have been drawn to frequentist statistics by Correll & Gleicher (2014), who compared results to Cohen’s D , and to Bayesian statistics by Kim et al. (2019), who evaluated plots based on their impact on the users’ prior beliefs. The comparison to hypothesis testing is far more natural for the lineup protocol, which has a visual test statistic (Majumder et al. 2013) and can be compared to standard statistical tests using power curves (Li et al. 2024). The connections between lineups and uncertainty visualisation are numerous and have been previously identified in the development of HOPS (Hullman et al. 2015).



The lineup protocol and uncertainty visualisations are similar: lineups were designed for checking if perceived patterns are real or merely the result of chance (Buja et al. 2009, Wickham et al. 2010). As both approaches are attempting to do the same thing, it is likely that we are unable to leverage the lineup protocol directly to evaluate uncertainty visualisation, but a new evaluation methodology should try to learn from the success of the lineup approach. Designing an implicit testing method for uncertainty visualisation that allows us to draw parallels to standard notions of statistical significance would solve many of the issues with the current evaluation approaches.

5. Conclusions and Future Work

This paper examines the literature and provides suggestions for a structural framework to support uncertainty visualisation. Particularly, we propose that uncertainty visualisation should accomplish signal suppression, dampening weak signals and amplifying strong signals. We have also highlighted several gaps in the existing literature.



Experimental practices on uncertainty visualisation need standards. Some existing evaluation experiments treat uncertainty as a signal, while others treat uncertainty as noise. As a result, it is difficult to combine results from papers to get a meaningful sense of how uncertainty information is understood by a viewer. Researchers need to ensure that when they identify the motivation behind their visualisation technique, their evaluation methods align with the stated goals of the paper.

Experimental methods that evaluate uncertainty as noise need to be developed. Research

into separability and integrability of signal and noise is of particular interest to uncertainty visualisation, as it allows assessment of the interference between the two. When designing experiments, authors often choose aesthetics that are visually distinguishable; uncertainty visualisation authors should consider doing the opposite.

Uncertainty needs to be formalised within the grammar of graphics. Some of this formalisation was done by Kay (2023), but it focuses only on the visualisation of univariate distributions. Giving authors the ability to describe uncertainty visualisations in terms of statistics, geometries and aesthetics will support evaluation experiments that can build towards a cohesive theory of visualising uncertainty.

Software that allows users to easily perform signal suppression is needed. Existing uncertainty visualisation methods view a distribution as its own object, and there are no software options treating “an uncertainty visualisation as a function of an existing visualisation” philosophy.

Signal suppression is an undeveloped area of visualisation research, and developing methods for the practice may require us to challenge our entire notion of what makes a good visualisation.

Reproducibility

The R packages were used for this work were: `tidyverse` (Wickham et al. 2019), `RColorBrewer` (Neuwirth 2022), `scales` (Wickham et al. 2023), `sf` (Pebesma & Bivand 2023), `urbanmapr` (Strochak et al. 2024), `flextable` (Gohel & Skintzos 2024), `colorspace` (Stauffer et al. 2009), `ggdist` (Kay 2023), `ggdibbler` (Mason et al. 2026), `patchwork` (Pedersen 2025), `distributional` (O’Hara-Wild et al. 2024), `ggthemes` (Arnold 2024), `broom` (Robinson et al. 2026), and `rgeos` (Bivand & Rundel 2023). The GitHub repository for this paper can be found at <https://github.com/harriet-mason/ARSA-UncertaintyLitReview>, which contains the files required to reproduce this article in full.

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